Information about the course

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Building 7 (Leonardo Campus), DEIB,
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Lecture timetable:

Monday, 15.15-18.15 –Lesson (T.1.3, 300)

Tuesday, 16.15-19.15 – Exercise/Lab (8.0.1, 260)

Wednesday, 8.15-10.15 –Lesson (T.2.2, 312)

Books: - Lesson notes on WeBeep website and on castellidezza.faculty.polimi.it

- N. Mohan, T.M. Undeland, W.P. Robbins, «Power Electronics: Converters, Applications, and Design», Editore: John Wiley & Sons

Consultation timetable: by appointment (Leonardo or Bovisa campus)

Course Objective

The main objective of the course consists of the definition of the procedures for the **design** of an **Electrical Drive**.

To achieve this objective, different configurations of **electrical machines**, **power supplies**, **control** systems are presented.

Table of Contents

Lectures program

- Principle of electrical engineering and energy conversion
- DC motor: steady-state and dynamic model, static converter, flux, speed and torque control schemes.
- Induction motor: steady-state equivalent circuits; 4 and 5 parameters dynamic model, voltage and current source inverters, state observers, field oriented control, scalar control scheme, DSC and DTC.
- AC Brushless: dynamic model, field oriented control. Synchronous Reluctance machine.
- DC Brushless: static converter and control scheme.
- Switched reluctance machine (SRM): static converter and control scheme.
- Industrial Case studies

Computer laboratory and experimental activities:

- Design of the control of a dc machine
- Design of the field oriented control of an induction machine
- Design of the scalar control of an induction machine
- Design of the direct torque control of an induction machine
- Design of the field oriented control of an ac brushless

Course assessment

The **exam** consists of an **oral test** (in Italian or English) to verify the knowledge of the electrical drives and a discussion on a **report** of one of the exercise activities, taken during the semester.

The exam consists of two main questions on **two different electrical drives** in order to test the knowledge of the student about:

- dynamic models of the electrical machine, starting from its structure
- expression of the electromagnetic torque
- control schemes (mainly torque and flux control loops)
- operating regions
- estimators (eventually)

The **report** has to show the definition of the parameters of the control system and the results of the simulation of the application (Matlab/Simulink).

Course assessment

The **exam** is organized by **appointment**.
When you think you are prepared, send me an email ten days before your preferred date.
If you want to take the exam on the **official session**, I ask you to send an email in the same way.

I will use Kahoot or Socrative (FRANCESCO 4412)

Preparatory tutoring calendar

BASICS OF ELECTRICAL MACHINES

Giorno	Orario	Room	Argomento/Subject	Tutor
22/02/2023	17:15-20:15	T0.3	Circuiti magnetici/ Magnetic circuits	Marzio Barresi
23/02/2023	17:15-20:15	T0.3	Trasformatore/Transformer	Marzio Barresi
24/02/2023	17:15-20:15	2.1.3	Macchina in CC/DC Machine	Marzio Barresi
01/03/2023	17:15-20:15	2.1.3	Campo magnetico rotante/Rotating magnetic field	Marzio Barresi
02/03/2023	17:15-20:15		Macchina sincrona/Synchronous machine	Marzio Barresi
03/03/2023	17:15-20:15		Macchina asincrona/Induction machine	Marzio Barresi
08/03/2023	17:15-19:15		Macchina asincrona/Induction machine	Marzio Barresi

Symbols

Abbreviation	*Variable*	*Unit*
v	voltage	V
i	current	A
$e(t)$	induced voltage e.m.f.	V
R	resistance	Ω
L / C	inductance/capacitance	H / F
$\mathcal{R}$ / Λ	reluctance/permeance	H^{-1} / H
φ	magnetic flux	Wb
ψ	flux linkage	Wb
f	force	N
T	torque	Nm
$\mathcal{V}$	linear speed	m/s
ω	angular speed/frequency	rad/s
J	inertia	kg m^2
n_p	number of pole pairs	